✅ PRACTICAL 1 – Data Preprocessing

1. **What is data preprocessing?**
   Data preprocessing is the process of cleaning and transforming raw data into a suitable format for analysis. It includes handling missing values, encoding categorical variables, and scaling features.
2. **Why is feature scaling required in machine learning?**
   Feature scaling ensures that all features have similar ranges so no variable dominates others. It is especially important for distance-based algorithms like k-NN and gradient-based methods.
3. **What is the difference between normalization and standardization?**
   Normalization scales data to a fixed range, usually between 0 and 1. Standardization transforms data to have a mean of 0 and standard deviation of 1.
4. **What is feature engineering?**
   Feature engineering involves creating new features or modifying existing ones to improve model performance. It helps in capturing hidden patterns in data.
5. **Name any two techniques used to handle missing data.**
   Missing data can be handled using mean imputation for numerical data and mode imputation for categorical data. Another approach is removing rows or columns with too many missing values.

✅ PRACTICAL 2 – EDA

1. **What is EDA?**
   Exploratory Data Analysis is the process of analyzing datasets using statistics and visualizations. It helps understand patterns, relationships, and anomalies in data.
2. **What is correlation?**
   Correlation measures the strength and direction of the relationship between two variables. It ranges from -1 (negative) to +1 (positive).
3. **What is covariance?**
   Covariance indicates how two variables vary together. A positive value shows they increase together, while a negative value shows opposite behavior.
4. **What is the range of correlation coefficient?**
   The correlation coefficient ranges from -1 to +1. Values close to ±1 indicate strong relationships, while 0 indicates no relationship.

5. **Name one visualization technique used in EDA.**
   A common visualization technique is a histogram, which shows the distribution of numerical data. Other examples include boxplots and scatter plots.

✅ PRACTICAL 3 – Logistic Regression

1. **What type of problem does Logistic Regression solve?**
   Logistic Regression is used for classification problems, especially binary classification. It predicts the probability of a class label.
2. **What is the range of the sigmoid function?**
   The sigmoid function outputs values between 0 and 1. It converts linear output into probability values.
3. **What does accuracy represent?**
   Accuracy represents the proportion of correctly predicted instances out of total predictions. It is a basic performance metric for classification.
4. **Name one evaluation metric for classification.**
   One evaluation metric is precision, which measures how many predicted positives are actually correct. Others include recall and F1-score.
5. **Is logistic regression a linear or non-linear model?**
   Logistic regression is a linear model because it uses a linear combination of features. The non-linearity comes from the sigmoid function.

✅ PRACTICAL 4 – k-NN

1. **Is k-NN a parametric or non-parametric algorithm?**
   k-NN is a non-parametric algorithm because it does not assume any underlying data distribution. It directly uses training data during prediction.
2. **What does the value of k represent?**
   The value of k represents the number of nearest neighbors considered for making predictions. It affects model complexity.
3. **Why is k-NN called a lazy learner?**
   k-NN is called a lazy learner because it does not train a model in advance. It stores data and performs computation during prediction.
4. **Which distance metric is commonly used in k-NN?**
   Euclidean distance is the most commonly used metric. Other options include Manhattan and Minkowski distance.

5. **What happens when k is very large?**
   When k is very large, the model becomes too generalized. This leads to underfitting and loss of important patterns.

✅ PRACTICAL 5 – Regression

1. **What is regression in machine learning?**
   Regression is used to predict continuous numerical values. It models the relationship between independent and dependent variables.
2. **What is the difference between simple and multiple linear regression?**
   Simple linear regression uses one independent variable, while multiple linear regression uses two or more variables.
3. **What is polynomial regression?**
   Polynomial regression models nonlinear relationships by adding polynomial terms. It is an extension of linear regression.
4. **Why is polynomial regression considered linear in parameters?**
   Although it uses powers of variables, the coefficients remain linear. Hence it is linear in terms of parameters.
5. **What is overfitting in polynomial regression?**
   Overfitting occurs when the model fits noise instead of actual patterns. It performs well on training data but poorly on new data.
6. **What is the difference between MSE and RMSE?**
   MSE is the average of squared errors, while RMSE is its square root. RMSE is in the same units as the target variable.
7. **What is R^2 score and why is it important?**
   R^2 score measures how well the model explains variance in data. A higher value indicates better performance.
8. **How do you decide the degree of the polynomial?**
   The degree is chosen using validation or cross-validation. It balances bias and variance.
9. **Which model is better for nonlinear data and why?**
   Polynomial regression is better for nonlinear data because it captures curved relationships.
10. **Can polynomial regression be used with multiple input features?**
    Yes, polynomial regression can handle multiple features. Interaction terms can also be included.

✅ PRACTICAL 6 – Regularization

1. **What is regularization?**
   Regularization is a technique that adds a penalty to model coefficients. It prevents overfitting.
2. **Why is regularization used in regression?**
   It reduces model complexity and improves generalization. It helps avoid overfitting.
3. **What is the difference between L1 and L2 regularization?**
   L1 (Lasso) can shrink some coefficients to zero, while L2 (Ridge) reduces coefficients but keeps all features.
4. **Why can Lasso perform feature selection?**
   Lasso forces some coefficients to exactly zero. This effectively removes irrelevant features.
5. **What is the role of alpha (λ)?**
   Alpha controls the strength of regularization. A higher value increases penalty.
6. **Why should features be scaled before Ridge/Lasso?**
   Scaling ensures all features are treated equally. Otherwise, larger scale features dominate.
7. **What is overfitting?**
   Overfitting happens when a model learns noise along with data. It reduces performance on new data.
8. **What is validation MSE?**
   Validation MSE is the error measured on unseen validation data. It helps in model selection.

✅ PRACTICAL 7 – Model Evaluation

1. **What is model evaluation?**
   Model evaluation measures how well a model performs on unseen data. It helps compare models.
2. **What is the difference between training error and validation error?**
   Training error is calculated on training data, while validation error is calculated on unseen data.
3. **What is cross-validation?**
   Cross-validation splits data into multiple folds and evaluates model multiple times. It gives reliable performance estimates.

4. **Why is k-fold cross-validation preferred over a single split?**
   It reduces dependency on a single split and gives more stable results.
5. **What is MSE?**
   Mean Squared Error is the average squared difference between actual and predicted values.
6. **What is hyperparameter tuning?**
   It is the process of selecting the best parameters like alpha or k. It improves model performance.
7. **Why can a single validation split be misleading?**
   Results may depend on how data is split. It may not represent overall performance.
8. **What is the purpose of choosing the best alpha?**
   The best alpha balances bias and variance. It minimizes validation error.

✅ PRACTICAL 8 – Bias-Variance

1. **What is bias in machine learning?**
   Bias is error due to overly simple assumptions. High bias leads to underfitting.
2. **What is variance in machine learning?**
   Variance is error due to sensitivity to training data. High variance leads to overfitting.
3. **What is underfitting?**
   Underfitting occurs when the model is too simple. It cannot capture patterns.
4. **What is overfitting?**
   Overfitting occurs when the model learns noise. It performs poorly on new data.
5. **Why does a high-degree polynomial overfit?**
   It becomes too flexible and fits noise in data. This increases variance.
6. **Why does a low-degree polynomial underfit?**
   It is too simple to capture patterns. This increases bias.
7. **What is the bias–variance trade-off?**
   It is the balance between model simplicity and complexity. The goal is optimal performance.
8. **How can regularization reduce variance?**
   Regularization simplifies the model by shrinking coefficients. This reduces overfitting.

✅ PRACTICAL 9 – Perceptron

1. **What is a perceptron?**
   A perceptron is a single-layer neural network used for binary classification. It creates a linear decision boundary.
2. **Why is a single-layer perceptron used only for linear classification?**
   It can only separate linearly separable data. It cannot model complex patterns.
3. **What is the role of the activation function?**
   It determines whether a neuron should activate. It introduces non-linearity in decision making.
4. **What is the perceptron learning rule?**
   Weights are updated based on prediction error. This improves classification accuracy.
5. **Why can a perceptron not solve XOR?**
   XOR is not linearly separable. A single line cannot separate the classes.
6. **What is the difference between weights and bias?**
   Weights determine feature importance. Bias shifts the decision boundary.
7. **What is binary classification?**
   Binary classification involves classifying data into two categories. Example: spam or not spam.

✅ PRACTICAL 10 – MLP

1. **What is a Multilayer Perceptron?**
   It is a neural network with multiple layers including hidden layers. It can model complex nonlinear relationships.
2. **What is the role of hidden layers?**
   Hidden layers extract features and learn patterns. They enable deep learning.
3. **What is backpropagation?**
   It is an algorithm that updates weights using error gradients. It minimizes loss.
4. **Why is MLP better than a single-layer perceptron?**
   MLP can solve nonlinear problems like XOR. It is more powerful.
5. **What are activation functions?**
   Functions like ReLU and sigmoid introduce non-linearity. They help neural networks learn complex patterns.
6. **What is gradient descent?**
   It is an optimization technique used to minimize loss. It updates weights iteratively.

7. **What is the difference between forward propagation and backpropagation?**
   Forward propagation computes outputs. Backpropagation updates weights based on errors.
8. **Why is feature scaling important in MLP?**
   Scaling improves convergence speed and stability. It prevents large values from dominating.

🎯 ADDITIONAL EXTRA VIVA QUESTIONS

🔹 Data Preprocessing & Encoding

1. **What is the difference between dropping and imputing missing values?**
   Dropping removes rows/columns with missing data, which may reduce dataset size. Imputation fills missing values, preserving data but may introduce bias.
2. **When should you remove a feature instead of imputing it?**
   A feature should be removed when it has too many missing values or low relevance. This prevents noise and improves model performance.
3. **What problem does label encoding create?**
   Label encoding may introduce false ordinal relationships between categories. Models may assume one category is greater than another.
4. **Why is one-hot encoding preferred over label encoding?**
   One-hot encoding avoids ordinal assumptions by creating separate binary columns. It is suitable for nominal data.
5. **What is dummy variable trap?**
   It occurs when one-hot encoded variables are highly correlated. It can be avoided by dropping one column.

🔹 Feature Scaling

6. **What is Min-Max scaling?**
   It rescales data to a fixed range, usually between 0 and 1. It preserves the shape of the distribution.
7. **When should you use StandardScaler over MinMaxScaler?**
   StandardScaler is preferred when data has outliers. It centers data around mean and reduces effect of extreme values.
8. **Does scaling affect tree-based models?**
   No, tree-based models like Decision Trees are not affected by scaling. They split based on feature values, not distances.

EDA & Visualization

9. **What is the purpose of a histogram?**
   A histogram shows the distribution of numerical data. It helps identify skewness and spread.
10. **What is the purpose of a pairplot?**
    Pairplot shows relationships between multiple variables. It helps visualize correlations and patterns.
11. **How do you identify outliers using boxplot?**
    Outliers are points beyond whiskers of the boxplot. They lie outside 1.5×IQR range.
12. **What is multicollinearity?**
    It occurs when independent variables are highly correlated. It can affect regression model performance.
13. **Why is correlation matrix important?**
    It shows relationships between variables. Helps in feature selection and detecting redundancy.

Logistic Regression

14. **What is a confusion matrix?**
    It is a table showing prediction results: TP, TN, FP, FN. It helps evaluate classification performance.
15. **What is recall?**
    Recall measures how many actual positives are correctly predicted. It focuses on minimizing false negatives.
16. **What is F1-score?**
    F1-score is the harmonic mean of precision and recall. It balances both metrics.
17. **Why compare models with and without scaling in logistic regression?**
    Scaling can improve convergence and accuracy. It ensures features contribute equally.

k-NN

18. **Why is k-NN sensitive to noise?**
    k-NN relies on nearest neighbors, so noisy data can mislead predictions. It directly affects accuracy.

19. **How do you choose optimal k?**
    Optimal k is chosen using validation or cross-validation. It balances bias and variance.
20. **What is the time complexity of k-NN?**
    k-NN has high prediction time because it compares with all training data. It is computationally expensive.

- **Regression Models**

21. **What assumptions does linear regression make?**
    It assumes linearity, independence, homoscedasticity, and normal distribution of errors. Violations affect performance.
22. **What is residual error?**
    Residual is the difference between actual and predicted values. It indicates model accuracy.
23. **What is a regression curve?**
    It is a line or curve showing relationship between variables. It helps visualize model fit.
24. **Why compare linear and polynomial regression?**
    To check if nonlinear patterns exist. Polynomial may fit better if data is not linear.

- **Regularization**

25. **What happens if alpha is too high?**
    Model becomes too simple and underfits. Important features may be ignored.
26. **What happens if alpha is too low?**
    Regularization effect is weak, leading to overfitting. Model may memorize noise.
27. **Why compare Ridge and Lasso coefficients?**
    It helps understand feature importance and selection. Lasso may eliminate features.

Model Evaluation & Cross Validation

28. **What is cross-validation score?**
    It is the average performance across folds. It gives a reliable estimate of model performance.
29. **Why is cross-validation better than train-test split?**
    It uses multiple splits, reducing randomness. Results are more stable.
30. **What is validation curve?**
    It shows model performance for different hyperparameter values. Helps in tuning.
31. **What is overfitting vs underfitting in evaluation?**
    Overfitting → low training error, high validation error.
    Underfitting → high error in both.

Model Selection

32. **How do you select the best model?**
    By comparing performance metrics like MSE or accuracy. The model with lowest error is preferred.
33. **Why is lowest MSE important?**
    Lower MSE indicates better predictions. It means smaller errors.
34. **Why compare validation and cross-validation results?**
    To ensure model consistency. Large differences indicate instability.

Advanced Practical Understanding

35. **What is feature importance?**
    It shows how much each feature contributes to prediction. Helps in feature selection.
36. **What is pipeline in machine learning?**
    A pipeline automates preprocessing and modeling steps. It ensures reproducibility.
37. **Why is reproducibility important?**
    It ensures consistent results across runs. Controlled using random_state.
38. **What is random_state?**
    It controls randomness in data splitting or models. Ensures same output every time.

39. **What is data leakage?**
    It occurs when training data contains information from test data. It leads to overly optimistic results.
40. **Why should scaling be done after train-test split?**
    Scaling before split leaks information. It should be done only on training data.